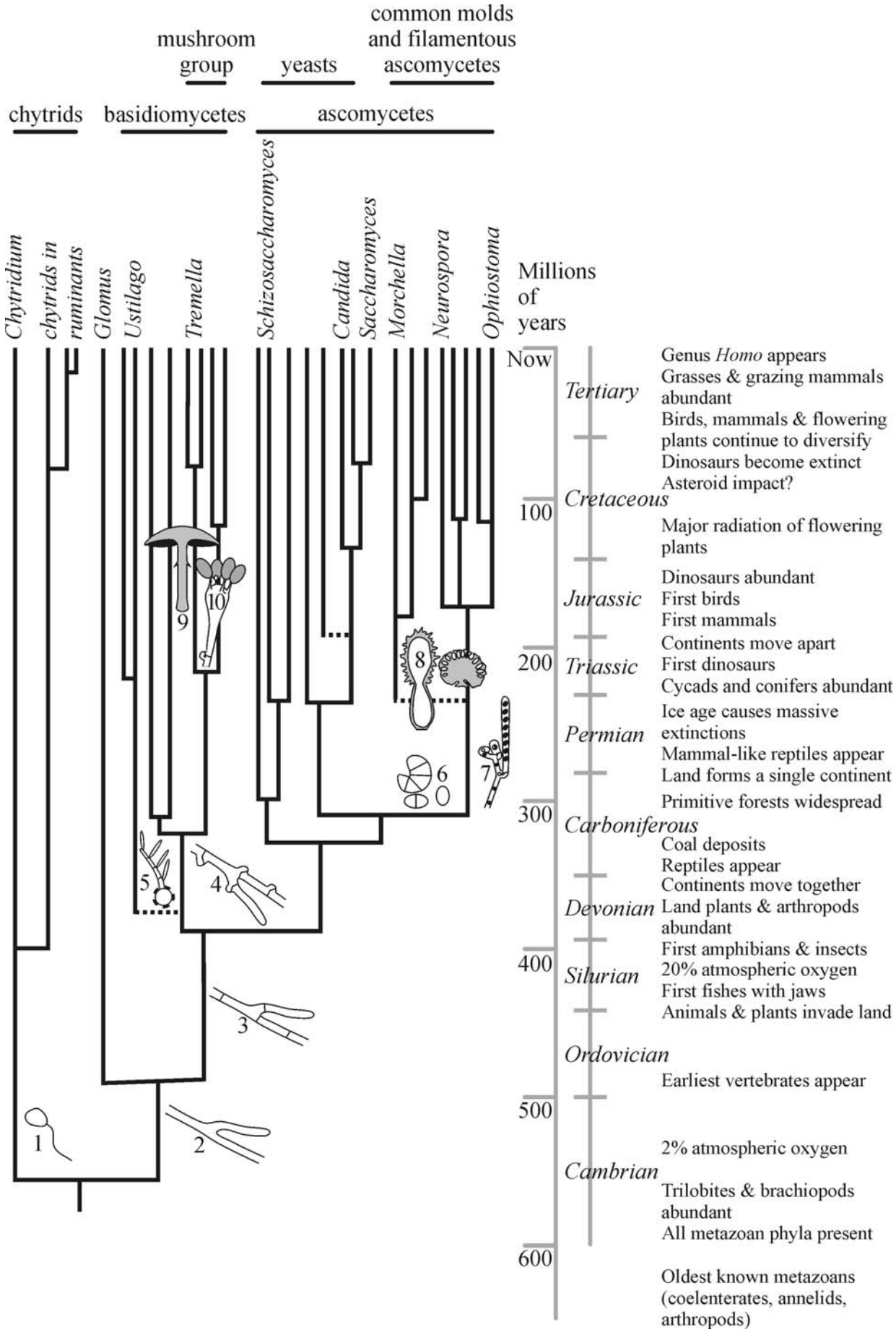

common molds and filamentous ascomycetes
mushroom group
yeasts
chytrids
basidiomycetes
ascomycetes
Chytridium
chytrids in ruminants
Glomus
Ustilago
Tremella
Schizosaccharomyces
Candida
Saccharomyces
Morchella
Neurospora
Ophiostoma
Millions of years
Now
100
200
300
400
500
600
Tertiary
Cretaceous
Jurassic
Triassic
Permian
Carboniferous
Devonian
Silurian
Ordovician
Cambrian
Genus Homo appears
Grasses & grazing mammals abundant
Birds, mammals & flowering plants continue to diversify
Dinosaurs become extinct
Asteroid impact?
Major radiation of flowering plants
Dinosaurs abundant
First birds
First mammals
Continents move apart
First dinosaurs
Cycads and conifers abundant
Ice age causes massive extinctions
Mammal-like reptiles appear
Land forms a single continent
Primitive forests widespread
Coal deposits
Reptiles appear
Continents move together
Land plants & arthropods abundant
First amphibians & insects
20% atmospheric oxygen
First fishes with jaws
Animals & plants invade land
Earliest vertebrates appear
2% atmospheric oxygen
Trilobites & brachiopods abundant
All metazoan phyla present
Oldest known metazoans (coelenterates, annelids, arthropods)
1
2
3
4
5
6
7
8
9
10

A summary of fungal evolution

This is a cladogram showing phylogeny of the true fungi based on the 18 S rDNA gene sequence. Branch lengths in the cladogram are proportional to the average rate of nucleotide substitution (1% per 100 million years), so the cladogram becomes an evolutionary tree, which has been calibrated using fossil fungi, fungal hosts and/or symbionts. The geological time timescale on the right shows the context of other major evolutionary events in geological time. The numerals and cartoons on the cladogram illustrate major milestones of fungal morphological evolution: terrestrial higher fungi diverged from water moulds (**1**) as branching filaments without septa (**2**) about 550 million years ago (Mya); the Glomaceae diverged from the progenitor of ascomycetes and basidiomycetes about 490 Mya, and the latter lineage evolved septate filaments (**3**); clamp connections mark early basidiomycetes (**4**); basidia (smut-like, **5**), asexual spores (**6**) and asci (**7**) probably evolved early in the major radiations of basidiomycetes and ascomycetes; filamentous ascomycetes diverged from the yeast lineage about 310 Mya, and fruiting bodies (**8**) presumably evolved before the Permian divergences because they are present in all the lineages today; mushroom fungi (**9**), with their characteristic holobasidium (**10**) probably radiated 130-200 Mya, soon after flowering plants became an important part of the flora.

It's interesting that coals deposited in the Cretaceous and Tertiary periods show much more evidence of fungal decay than the much older Carboniferous coals, reflecting the radiation of aggressive wood decay basidiomycetes from the Triassic onwards.

Also note the relatively recent radiation of (anaerobic) chytrids as grasses and grazing mammals became more abundant.

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